

A computational toolkit for assessing phonological specification in phonetic data: Discrete Cosine Transform, Micro-Prosodic Sampling, Bayesian Classification

Abstract

Many previous studies have argued that some phonological elements lack phonetic targets. In these cases, the phonetic signal is structured by the segments surrounding the targetless item, or, more specifically, by a linear interpolation from segments preceding the targetless element to segments following the targetless element. The question addressed in this paper is, given that natural variability is pervasive in phonetic data, what degree of linear interpolation in the phonetics justifies us to conclude that a phonological element lacks a target? We present a new computational toolkit consisting of Discrete Cosine Transform (DCT), Micro-prosodic sampling and Bayesian classification, for rigorous evaluation of linear interpolation on the basis of phonetic data. First, we use DCT to compress high dimensional phonetic data into cosine components that correspond to linguistically meaningful signal modulations. Micro-prosodic sampling of DCT coefficients allows us to simulate targetless segments as linear interpolations with natural levels of phonetic variability. These two computational tools allow us to express competing phonological hypotheses (target present vs. target absent) in the dimensions of the phonetic data. We next use Bayesian classification to assign probabilities of targetlessness to new data on a token-by-token basis. The empirical material used to demonstrate the approach comes from Electromagnetic Articulography recordings of high vowel devoicing in Japanese, although our computational toolkit can easily be applied to other cases. Most generally put, we maintain that our approach instantiates a happy marriage between computational methodologies with actual phonetic data, from the viewpoint of abstract phonological theory.

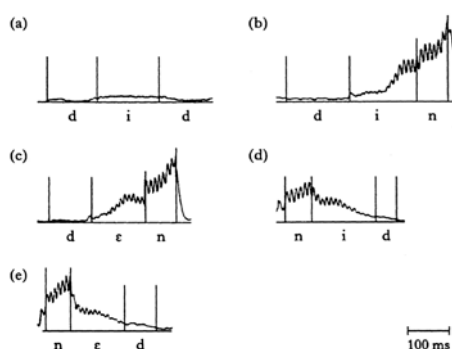
1.0 Introduction

1.1 Linear interpolation as evidence for phonological underspecification

The assumption in early generative phonology that every segment is specified for every distinctive feature and therefore receives “a phonetic command” for all the phonetic dimensions represented by distinctive features (e.g., Chomsky and Halle 1968:403-419) has given way to various proposals regarding underspecification (Archangeli 1988; Keating 1988). Building on the phonological theory of feature underspecification, Keating (1988) observed that some segments lack a particular “phonetic target” in some dimension. One example is English /h/, which on spectrograms can look like a linear interpolation from the preceding segment to the following segment.

The idea of phonetic underspecification—or in more neutral terms, apparent phonetic “targetlessness”—plays a central role in many phonological analyses. In the realm of intonation, Pierrehumbert (1980) argues that modeling intonational contours of English can be best achieved by only sparsely specifying H(igh) and L(ow) targets, rather than specifying all syllables in an utterance for tone. Sparse tonal specification has been extended to many languages (e.g., Beckman and Pierrehumbert 1986; Myers 1998) and now constitutes a fundamental assumption in the Autosegmental Metrical (AM) theory of intonation (Jun 2014; Arvaniti and Ladd 2015; c.f., Xu et al. 2015). Another oft-discussed example is the comparison of nasality in English and French, developed by Cohn (1993). She compared the phonetics of nasalization in French, where nasalization is contrastive in vowels, and English, where nasalization is not contrastive in vowels. Figure 1 reproduces the English data. On the basis of these measurements, she argues that in English, vowel nasalization before a nasal consonant involves an interpolation from [-nasal] to [+nasal], with the vowel being underspecified for [nasal].

Figure 1
Nasal flow patterns in English, based on Cohn (1993: 60).



Linear interpolation is indeed the crux of many phonetic arguments for phonological underspecification. Consider an ABC sequence, where the phonological specification of B is at issue. Whether observed in the domain of intonation (Pierrehumbert and Beckman 1988: 37-38), vowels (Browman and Goldstein 1992) or consonants (Cohn 1993; Keating 1988) linear interpolation on some phonetic dimension between A and C constitutes an argument for the “targetlessness” of some phonological feature of B. But there is one question that is not addressed sufficiently in the previous literature: how linear is linear?

Because actual articulation is never completely linear, how much deviation from pure linearity would justify us to reject the targetlessness hypothesis?¹ Consider again Figure 1. Does (b) really involve linear interpolation? Just looking at the contour, it looks like a quadratic function. Similarly, (c) looks as if there is a [+nasal] target in the middle of the vowel, or it involves a logarithmic function, because there is a plateau of nasality at the second phase of the vowel. On the other hand, perhaps these data are just noisy actuations of a pure linear trajectory. Since humans are biological organisms not speaking machines, phonetic data always come with some natural variability. How do we decide whether observed deviation from linearity is not due to such natural variability? In more general terms, taking the case of ABC again, how do we distinguish complete targetlessness of B from (heavy) reduction of B, for example? How do we know that B is truly targetless rather than just heavily susceptible to coarticulation with surrounding segments (c.f., Recasens and Espinosa 2009)?

In this paper, we introduce a computational toolkit to address these questions. There are three components. The first, Discrete Cosine Transform (DCT), we use to decompose high dimensional phonetic data into a low-dimensional frequency space that can be mapped to phonological form. In this space, we formulate competing phonological hypothesis, including the linear interpolation hypothesis. Next, we deploy stochastic sampling techniques developed for syllable micro-prosody (e.g., Shaw and Gafos, 2015). We estimate distributions over signal components in frequency space and sample from these distributions to convert competing phonological hypothesis into the real world spatial-temporal dimensions of the data. This step factors into the analysis the range of natural variability found in the phonetic data, allowing us to generate realistically noisy phonetic signals from competing phonological hypothesis. Finally, we train a Bayesian classifier on competing phonological hypotheses and use it to compute, on a token-to-token basis, the probability of linear interpolation given the phonetic data. Taken together, the suite of computational tools allows rigorous assessment

¹ Not all work on phonetic underspecification assumes linearity. Pierrehumbert (1980: 71), cited and discussed in Myers (1998: 368), argues that for H* accents in English “the F0 falls until it is time to start aiming for the next H* level”.

of the “targetlessness” claim on the basis of phonetic data. Most generally put, we maintain that our toolkit instantiates a happy marriage between computational methodologies with actual phonetic data, from the viewpoint of abstract phonological theory.

1.2 Japanese high vowel devoicing

To illustrate our computational approach, we examine high vowel devoicing in Tokyo Japanese (Fujimoto 2015; Kondo 2005; Tsuchida 1997). A classic description of this phenomenon is that “high vowels are devoiced between two voiceless consonants and after a voiceless consonant before a pause”. High vowel devoicing is well-studied, particularly in Tokyo Japanese, with research covering phonological conditions (e.g., Tsuchida 1997; Kondo 2005), acoustic and perceptual characteristics of devoiced vowels (Beckman and Shoji 1984; Faber and Vance 2000; Nielsen 2015), and articulatory studies of the vocal folds (Fujimoto et al. 2002; Hirose 1971). See Fujimoto (2015) for a recent overview. When vowels are devoiced, it is difficult to ascertain whether they maintain their lingual articulatory specification. While we now have a good understanding of many aspects of high vowel devoicing in Tokyo Japanese, the status of the lingual gestures of devoiced vowels is still poorly understood. Here, we entertain four hypotheses, stated in (1).

(1) Hypotheses about the status of lingual articulation in devoiced vowels

H1: **full lingual targets**—the lingual articulation of devoiced vowels is the same as for voiced counterparts.

H2: **reduced lingual targets**—the lingual articulation of devoiced vowels is phonetically reduced relative to voiced counterparts.

H3: **targetless**—devoiced vowels have no lingual articulatory target.

H4: **optionally targetless**—devoiced vowels are sometimes targetless

Not all previous studies are directly framed to support one of these hypotheses, but some studies are relevant to one of these hypotheses. Kawakami (1971: 24-26) argues that vowels delete in some environment (=H3) and devoice in others. Sometimes, vowels leave no trace of themselves but coarticulation on surrounding consonants, which has led some researchers to conclude that the vowels are deleted (Beckman and Shoji 1984; Beckman 1982; Whang 2014). If deletion is phonological, then the vowel should not exhibit a lingual gesture, predicting H3. Kondo (2001) argues that high vowel devoicing is actually deletion based on a phonological consideration. Devoicing in consecutive syllables is often prohibited, and Kondo argues that this prohibition stems from a constraint against complex onset or complex coda (i.e. *CCC). Even if vowel devoicing is phonological, some studies show that its application is optional or variable (Nielsen 2015; Fujimoto 2015), suggesting H4.

On the other hand, Kawahara (2015) argues that bimoraic foot-based truncation (Poser 1990) counts a voiceless vowel as one mora (e.g. [suto] from [sutoraiki] ‘strike’, *[stora]). If [u] was completely deleted losing its mora, the bimoraic truncation should result in *[stora]. This suggests that the lingual gesture of devoiced high vowels is phonologically present, and predict either H1 or H2. In particular, H1 is what would be predicted by a “gestural overlap theory” of high vowel devoicing (Faber and Vance 2000; Jun et al. 1998; Jun and Beckman 1993). In this theory, high vowel devoicing occurs when laryngeal abduction gestures of surrounding consonants overlap with the vowel. In this sense, high vowel devoicing processes in Japanese (and Korean) are “not...phonological rules, but as the result of extreme overlap and hiding of the vowel's glottal gesture by the consonant's gesture” (Jun and Beckman 1993: p.4). This passive devoicing hypothesis would predict that lingual gestures remain intact (=H1). Even if devoiced high vowels are not phonologically deleted or otherwise targetless, it would not be too surprising if the lingual gestures of high vowels are reduced. Due to devoicing, the acoustic consequences of a reduced lingual gesture would not be particularly audible. Hence from the standpoint of an effort-distinctiveness tradeoff, we expect reduction of oral gestures in high devoiced vowels (=H3).

The central question of this paper is how to distinguish between the hypotheses in (1). Especially, distinguishing between H2 and H3 is a specific case of the general issue raised in section 1.1. How do we know that a phonetic signal lacks a phonological target (=H3), rather than being reduced (=H2)? Although the empirical material used to demonstrate our approach comes from Japanese high vowel devoicing, the question that we are addressing is much more general: how do we assess linear interpolation in order to confirm the targetlessness of a particular segment?

A key tenet of our approach is to express abstract phonological hypotheses in the units of the phonetic data. Like snowflakes and fingerprints, no two phonetic signals are identical, even those that actuate identical phonological structures. This fact dictates that rigorous assessment of phonological hypotheses on the basis of phonetic data requires a probabilistic model of how phonological form maps to the phonetic signal. Following recent approaches to syllable micro-prosody (Shaw and Gafos 2015; Shaw et al. 2011; Shaw et al. 2009), we seek to estimate distributions that relate low dimensional phonological hypotheses to high dimensional phonetic data. Accordingly, we construct stochastic phonetic models that are parameterized by our phonological hypothesis as well as by the level of variability naturally present in the phonetic data. Before moving to the details of the computational toolkit, we briefly introduce the data to be modelled.

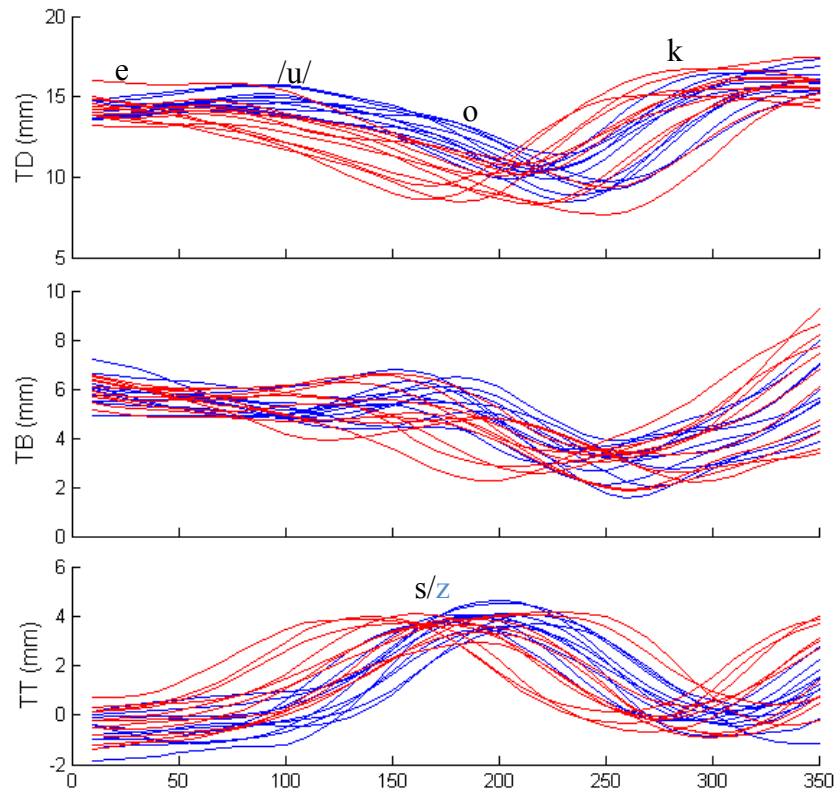
1.3 Electromagnetic Articulography recordings of Japanese vowels

The phonetic data used to illustrate our computational approach were drawn from a larger experiment using Electromagnetic Articulography (EMA) to track the movement of fleshpoints on the tongue during the production of voiced and voiceless vowels in Tokyo Japanese. Detailed experimental methods are reported in section 4. Here, we preview the raw data in order to make concrete the nature of the problem. Figure 2 provides representative data from one speaker producing 11 repetitions of the minimal pair / ϕ usoku/ ‘shortage’~/ ϕ uzoku/ ‘enclosurement’ in the carrier phrase /okee _____ to itte/ “Ok, say _____”. The movement trajectories span from the /e/ in /okee/ to the /k/ in / ϕ usoku/ (and in / ϕ uzoku/). In line with descriptions of high vowel devoicing in contemporary Tokyo Japanese, this speaker produced voiced /u/ in / ϕ uzoku/ and devoiced /u/ in / ϕ usoku/. Of interest for our case study is whether the lingual gesture of the devoiced vowel in / ϕ usoku/ has an articulatory target. The top panel of Figure 2 shows the height of the tongue dorsum (TD) sensor (y-axis) over time (x-axis) with / ϕ usoku/ (devoiced /u/) in red and / ϕ uzoku/ (voiced /u/) in blue. The middle panels show movement of the tongue blade (TB) and the bottom panel shows the tongue tip (TT). For the portion of the figure corresponding to /u/, the TD is lower for devoiced /u/ (/ ϕ usoku/, red lines) than for voiced /u/ (/ ϕ uzoku/, blue lines). At the very least, this pattern indicates that the devoiced vowel is phonetically reduced in this speaker’s pronunciation.

In the remainder of this paper, we describe a computational approach to evaluating the four hypotheses in (1) on the basis of continuous phonetic data, such as that shown in Figure 2.

Figure 2

Lingual articulatory trajectories from a female speaker of Tokyo Japanese producing / ϕ usoku/ (red lines) and / ϕ uzoku/ (blue lines).



2.0 Discrete Cosine Transform and Micro-prosodic Sampling

This section introduces the first two of three computational tools: Discrete Cosine Transform (DCT) and Micro-prosodic sampling. The purpose of these tools is to estimate distributions that characterize the phonetic data so that we can simulate realistically variable trajectories actuating phonological hypothesis (including the targetless hypothesis).

As a starting point, we assume that the speech articulators follow direct paths between articulatory goals (c.f., Browman and Goldstein 1992; Keating 1988). The idealized movement trajectory corresponding to the targetless vowel hypothesis (H3) would therefore be a linear trajectory from V_1 to V_3 (Lammert et al. 2014; Browman and Goldstein 1992). In real articulatory data, flesh-point trajectories are almost never straight lines. There are well-studied cases in which tongue trajectories are curved because of biomechanical factors even when the idealized movement based on phonological form would dictate a linear trajectory (Mooshammer et al. 1995; Perrier et al. 2003). To account for the numerous perturbations, biomechanical and otherwise, of linear trajectories between articulatory goals in speech production, we take a stochastic, data-driven approach, modelling actual trajectories as noisy actuations of phonological goals (Shaw and Davidson 2011; Shaw and Gafos 2010; Shaw et al. 2009).

Consider again the data in Figure 2. The trajectories begin with the vowel /e/ of the carrier phrase preceding the target words / ϕ usoku/ (red lines) and / ϕ uzoku/ (blue lines). The TD starts out high for the vowel /e/. The vowel /u/, if it is present, would follow the /e/. Some tokens show a slight rise in TD height at the start of the trajectory, which is expected if the tongue dorsum rises in height from /e/ to /u/; many tokens, however, particularly those of / ϕ usoku/, show a monotonic decrease in

height from /e/ to /o/, which is expected if there is no lingual target for /u/. The question we ask with the modelling is whether the observed trajectory from /e/ to /o/ is reliably different from a realistically variable linear trajectory between /e/ and /o/. If so, this would support the phonetic reduction hypothesis (H2); if not, the result would support the "targetless" hypothesis (H3), at least for some tokens (H4).

2.1 Discrete Cosine Transform

The TD trajectories in Figure 2 consist of 35 data points per trajectory, one data point for every 10 ms, due to the 100 Hz sampling rate used in the EMA recording (see section 4 for detailed experimental methods). The data points are not statistically independent. Rather, the height of the TD at any point in time, τ , is closely related to the height of the TD at earlier, $\tau-1$, or later time points, $\tau+1$. We use Discrete Cosine Transform (DCT), the first computational tool in our toolkit, to capture dependencies between data points. Doing so allows data compression and sparse representation, which both simplifies subsequent computation and facilitates generalization.

DCT represents the data as sums of cosines of different frequencies and amplitudes. In expressing spatial data in terms of harmonic components (i.e., frequency space), DCT is similar to Fast Fourier Transform (FFT) typically used to construct spectrograms from the acoustic signal. The main advantage of DCT in particular for our purpose here is that it represents the data with a small set of parameters, a general property of DCT (Jain 1989: 151), each of which has a clear linguistic interpretation. Also important is that DCT has a known inverse function, which we use to simulate TD trajectories from DCT components. Each cosine component of a DCT has an amplitude coefficient that is fitted to the data. We interpret the amplitude of the cosines as the degree to which a corresponding gesture modulates the TD trajectory. DCT has been used in some previous phonetic studies, which have shown that phonetic signals, particularly vowel formants, can be represented quite well with a small number of cosine components (Williams and Escudero 2014; Watson and Harrington 1999).

We provide a mathematical expression of DCT transform in (1) and represent it visually with an example from the data in Figure 3. In the numerical expression, $y(k)$ is the amplitude of the k^{th} cosine component. This is the output of the DCT. The other terms in the equation are as follows: L is the length of the trajectory (i.e., the number of data points); $x(n)$ is the trajectory of the data being modelled; $w(k) = \frac{1}{\sqrt{L}}$ when $k = 1$ and $w(k) = \sqrt{\frac{2}{L}}$ otherwise. The first DCT coefficient, $y(1)$, defines a straight line at a position above the average value of the data. This is because when $w(k) = 1$, the term of the cosine function is zero. This means that the first coefficient is equal to $\frac{\sum_{n=1}^L x(n)}{\sqrt{L}}$, the sum of all data points in the trajectory divided by the square root of the number of data points (c.f., $\frac{\sum_{n=1}^L x(n)}{L}$, the average trajectory). Each subsequent DCT component defines a cosine of increasing frequency, as increases to k linearly increase the term of the cosine function.

(1) Numerical expression of Discrete Cosine Transform

$$y(k) = w(k) \sum_{n=1}^L x(n) \cos\left(\frac{\pi(2n-1)(k-1)}{2L}\right) \quad k = 1, 2, \dots, L$$

where

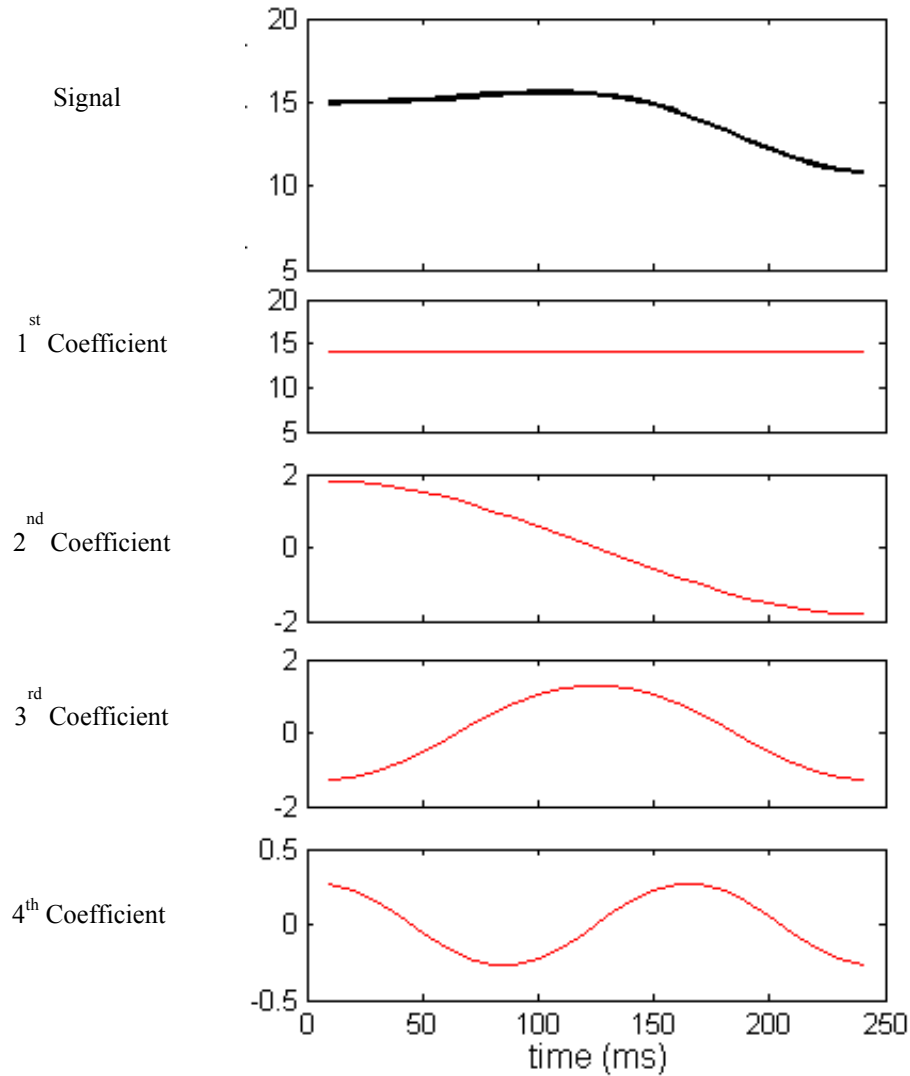
$$w(k) = \begin{cases} \frac{1}{\sqrt{L}} & k = 1 \\ \sqrt{\frac{2}{L}} & k = 2, \dots, L \end{cases}$$

$$\sqrt{\frac{2}{L}} \quad 2 \leq k \leq L$$

Figure 3 illustrates the DCT components of a TD trajectory. The top panel shows the trajectory, the vertical movement of the TD (y -axis) over time (x -axis). The first DCT coefficient defines a straight line at 14 mm (above the level of the occlusal plane). In the discussion below we refer to this line as the baseline TD height. Subsequent coefficients describe deviations from the line as cosine-shaped modulations of increasing frequency. These subsequent components, i.e., the second to the fourth cosine components, are centered on zero. The second coefficient captures the downward trend of the TD trajectory, ranging from +2 mm to -2 mm. Thus, the second coefficient captures the fact that, in this data, the TD starts high and then lowers over time and that the range of this lowering motion covers a 4 mm span. The third coefficient adds another modulation to the trajectory. Towards the middle of the trajectory there is a rise. The third coefficient indicates that this rise constitutes a modulation of the baseline trajectory on the order of ± 2 mm. The effect of the fourth cosine coefficient is more subtle, specifying modulations that are less than ± 0.5 mm.

Figure 3

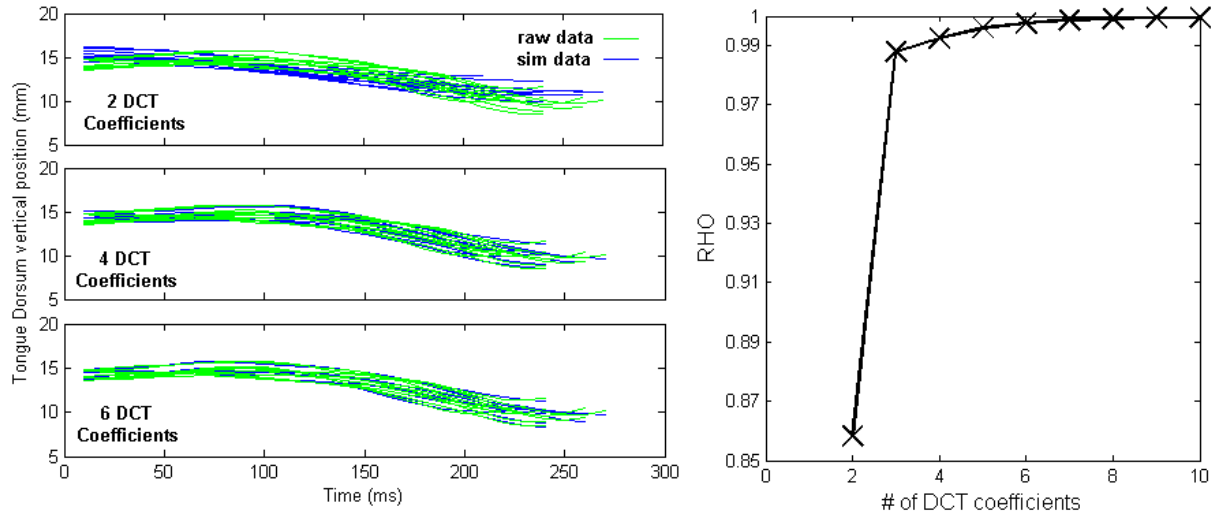
An illustration of Discrete Cosine Transfer (DCT) components for a TD trajectory. The top panel shows the signal. The bottom four panels show individual DCT components contributing to the signal.



To evaluate how many cosine components are needed to represent movement trajectories in the EMA data, we re-simulated the TD data shown in Figure 2 using different numbers of DCT coefficients and evaluated the degree of data loss as a function of the number of DCT coefficients. The number of DCT coefficients was varied from one to ten. The right panel of Figure 3 shows how the correlation between the raw data and the simulated data increases with the number of DCT coefficients. The correlation is $r = .858$ with just two cosine coefficients. Increasing the number of parameters to four increases the correlation up to $r = .992$. The correlation with six DCT coefficients is $r = .998$. The left side of Figure 4 illustrates the goodness of fit token by token. The same set of eleven / ϕ uzoku/ tokens from Figure 2 is re-displayed in green (in Figure 3, the tokens are truncated at the /o/). The blue lines show trajectories that were simulated from DCT coefficients. With four coefficients, the blue and the green trajectories overlap almost completely, illustrating nearly lossless compression.

Figure 4

The left panels show the raw data (green lines) and simulated trajectories (blue lines) for 11 tokens of / ϕ uzoku/. Simulations were based on two (top panel), four (middle panel) and six (bottom panel) DCT coefficients. The right panel shows the correlation between raw data and simulated data as a function of the number of DCT coefficients employed in data representation.

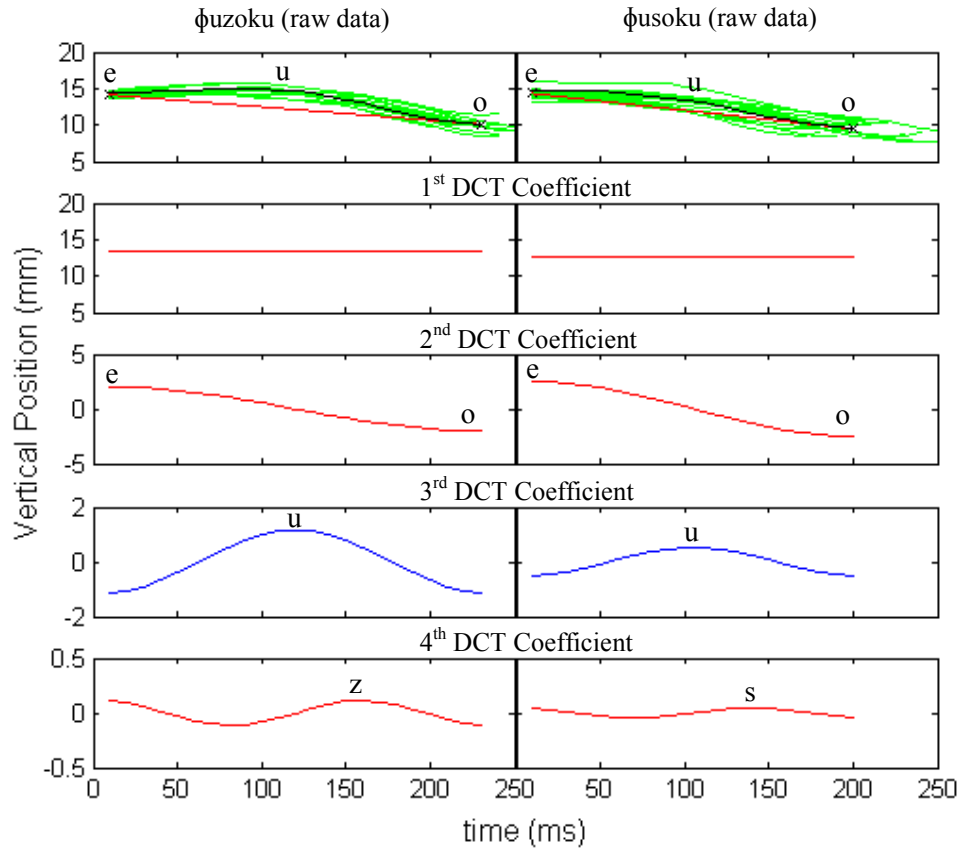


By using DCT, we can effectively reduce the dimensionality of the data without loss of precision. The results in Figure 3 indicate that 4 DCT coefficients are sufficient to retain a detailed phonetic representation of the TD signal. More importantly, the 4 DCT coefficients have clear linguistic interpretations on which we now elaborate.

Figure 5 re-displays the / ϕ usoku/~/ ϕ uzoku/ data from Figure 2 along with mean DCT components fit to the data. The left column shows the voiced /u/ in / ϕ uzoku/; the right column shows the devoiced /u/ in / ϕ usoku/. The top panels show the raw data (green lines) together with the average trajectory (black line) and a linear trajectory between /e/ and /o/ vowel (red line). The average trajectory was computed by averaging DCT coefficients. It can be observed that the average trajectory (black line) is closer to the linear trajectory for / ϕ usoku/ (right) than for / ϕ usoku/ (left). More to the point, each of the three cosines has a clear linguistic interpretation, which helps to isolate the difference in trajectory between voiced and voiceless vowels. The first cosine (3rd panel from the top) indicates a fall in TD height from /e/ to /o/. This component represents the vowel-to-vowel transition, which is similar for both words. Vowel to vowel intervals have long provided building blocks for speech production models and have a privileged theoretical status in some phonological models (Ohman 1966; Mrayati et al. 1988; Carré and Chennoukh 1995; Smith 1995). The second cosine represents an increase in TD height for the vowel /u/. This rise is present for both / ϕ usoku/ and / ϕ uzoku/ but the magnitude of the rise is greater for the voiced vowel in / ϕ uzoku/ than for the devoiced vowel in / ϕ usoku/. Thus, the third DCT coefficient isolates the difference between these words observed in Figure 2. Finally, the fourth DCT coefficient adds a subtle (< .5 mm) modulation to the TD trajectory. The time course of this modulation is consistent with coarticulatory effects of coronal consonants /s/ and /z/ on TD height.

Figure 5

Average DCT components for / ϕ uzoku/ (left) and / ϕ usoku/ (right). The raw data is displayed in the top panel. Average DCT components are plotted below.



To summarize, the first computational step in our approach is to express TD trajectories over /VCuCV/ sequences as the sum of four DCT components. The first of these components expresses a baseline TD position (akin to the intercept in regression analysis) while the second, third and fourth components capture linguistically meaningful modulation of vocal tract constrictions. We next make use of these compressed representations of phonetic detail to estimate distributions characterizing the phonetic expression of phonological form, including the targetless hypothesis, H3 in (1).

2.2 Micro-prosodic sampling

We refer to the next computational tool as “Micro-prosodic sampling” because it borrows from recent stochastic approaches to modelling prosodic structure in terms of gestural timing (Shaw and Gafos 2015; Shaw et al. 2011; Shaw and Gafos 2010; Shaw et al. 2009). These studies estimate distributions over spatio-temporally defined gestural landmarks (Gafos 2002) and sample from the distributions under different conditions. The parameters of such stochastic generators can be varied to test specific hypotheses about the phonological structure of the data, including the presence or absence of gestures (Shaw and Davidson 2011) or the syllabic affiliation of the segments (Shaw and Gafos 2015, 2010). We proceed here by defining Gaussian distributions over DCT coefficients instead of gestural landmarks.

Gestural landmarks and DCT coefficients both offer a sparse representation of detailed phonetic data. For the current case, an advantage of using DCT coefficients is that it is not necessary to parse specific gestural landmarks from the signal. Parsing gestural landmarks often relies on heuristic use of movement velocity profiles (Gafos et al. 2010; Shaw et al. 2009). In the trajectories

shown in Figure 2, however, it was not possible to identify clear velocity peaks corresponding to the different vowel gestures. Rather, the TD moved smoothly with more or less constant velocity from one vowel to the next, a pattern also reported in other kinematic data sets (Ohman 1966; e.g., Browman and Goldstein 1992). In data such as these, selecting a single point in time that corresponds to the vowel is arbitrary (Mücke et al. 2014). Our solution here is to model the entire trajectory, but in the compact and linguistically relevant form of DCT coefficients.

To formulate the targetless hypothesis in terms of DCT coefficients, we first fit a straight line from V_1 to V_3 in V_1CuCV_3 sequences. If there is no independent TD height target for /u/, i.e., H3 in (1), then we assume that the tongue dorsum position will follow a straight path from V_1 to V_3 . To formulate a stochastic version of this targetless trajectory, we coerced the linear interpolation between vowels into frequency space by fitting four DCT coefficients to the straight line from V_1 to V_3 . We then defined distributions over those DCT coefficients. The shape of the distributions was guided by analysis of the data. We chose Normal Gaussian distributions, since the DCT coefficients fit to our data did not significantly depart from Normality, according to Shapiro-Wilk tests. For the targetless hypothesis, the means of the distributions were the DCT coefficients fit to the linear interpolation between vowels. The standard deviation of the distributions was set to the standard deviation of DCT coefficients fit to the corresponding data. This ensures that we inject reasonable quantities of variation into the targetless trajectory. By expressing the interpolation line in the same stochastic four-parameter DCT space, we imbibe linear interpolation with variability along the same dimensions as variation in the data. Formalized as distributions over four DCT coefficients, corresponding to the bottom four panels of Figure 5, the targetless hypothesis thus has the same degrees of freedom as the vowel hypothesis, meaning that it varies in the same dimensions and to the same degree as the raw data that we model.

The computational method described above expresses the targetless hypothesis in the phonetic dimensions of the data, specifically the height of the tongue dorsum over time, as phonological control of the vocal tract passes from one vowel to the next. Table 1 provides a specific example. The top two rows show the DCT distributions of the raw data. The mean value of each coefficient is shown with standard deviation given in parenthesis. The bottom two rows provide the parameters for the targetless hypothesis for the same data. The mean parameters come from a four parameter DCT of the straight line trajectory, left-delimited by the mean target of V_1 and right-delimited by the mean target of V_3 . Note that the third coefficient, Co3, is zero, for the targetless hypothesis, indicating no rise from the trajectory defined by Co2 (see also Figure 5 for reference). The standard deviation of the targetless hypothesis is identical to the raw data because the level of variability in the targetless hypothesis is set to the level of variability in the data.

		Distributions over DCT coefficients			
		Co1	Co2	Co3	Co4
raw data	/φusoku/ (raw)	54.7(3.27)	7.18(1.80)	0.263(0.886)	0.879(0.565)
	/φuzoku/ (raw)	64.4(5.15)	6.25(2.31)	-1.44(1.95)	1.60(0.898)
targetless hypothesis	/φusoku/ (simulated)	53.3(3.27)	7.29(1.80)	0.00(0.886)	0.804(0.565)
	/φuzoku/ (simulated)	60.2(5.15)	7.96(2.31)	0.00(1.95)	0.880(0.898)

Table 1: mean and standard deviation (in parentheses) of DCT coefficients

Having defined distributions over DCT coefficients, we can sample from the DCT coefficients to simulate trajectories corresponding to the vowel deletion trajectory. The sampled DCT coefficients can then be used to specify the TD trajectory by applying the inverse DCT function to the coefficients. The formulae for simulating trajectories by applying inverse Discrete Cosine Transform (iDCT) is given in (2). As with the DCT expression in (1), L indicates the length of the trajectory; $x(n)$

is the trajectory, this time on the left side of the question; $y(k)$ represents the k^{th} DCT coefficient; and w is a constant. We simulated trajectories that were equal to the mean duration of the V_1 to V_3 signal with $k = 4$ DCT Coefficients.

(2) Numerical expression of inverse Discrete Cosine Transform

$$y(k) \sim N(\mu(k), \sigma(k))$$

$$x(n) = \sum_{k=1}^L w(k)y(k) \cos\left(\frac{\pi(2n-1)(k-1)}{2L}\right) \quad n = 1, 2, \dots, L$$

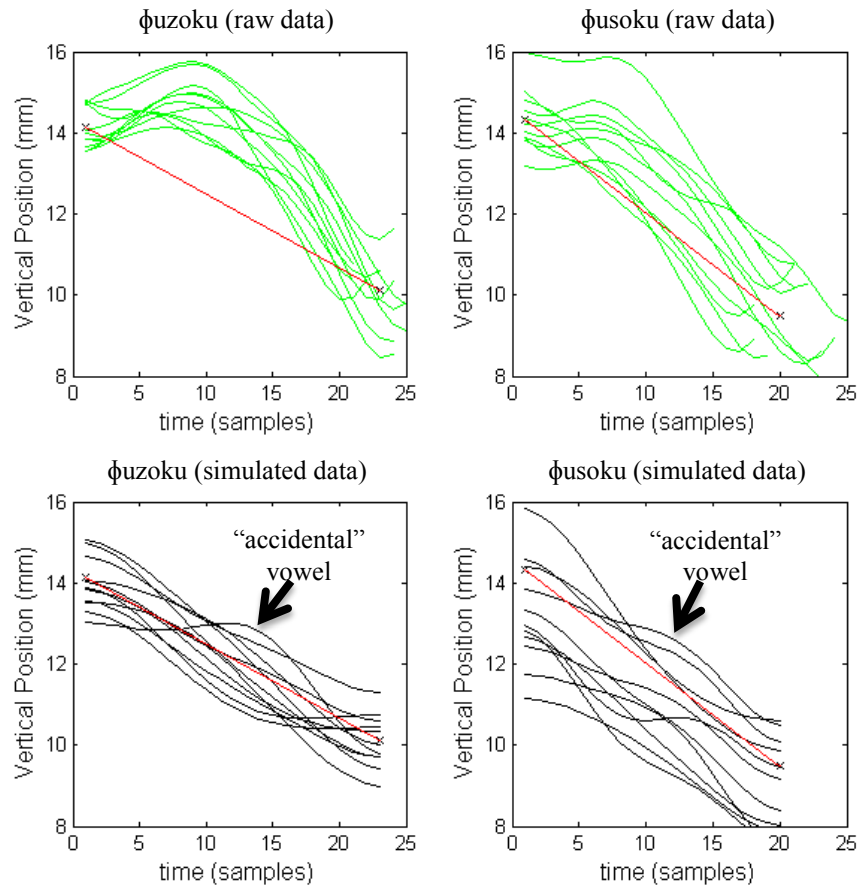
where

$$w(k) = \begin{cases} \frac{1}{\sqrt{L}} & k = 1 \\ \sqrt{\frac{2}{L}} & 2 \leq k \leq L \end{cases}$$

Figure 6 illustrates the simulations graphically. For reference, the top panels re-plot the data from Figure 2 in green lines. However, in this figure only the portion of the trajectory beginning with the /e/ from the carrier phrase and ending with the /o/ is shown. The TD trajectories from /ϕuzoku/ are shown on the left; the trajectories from /ϕusoku/ are on the right. The ‘x’'s denote vowel targets, the mean height of V_1 , /e/, and V_3 , /o/. The red line connects the means of the vowels and defines the direct vowel trajectory. Comparison of the left and right panels reveals that TD height in /ϕusoku/ tends to be closer to the line than does TD height in /ϕuzoku/, essentially the same observation we made in Figure 2 (but this time with reference to the linear trajectory). The bottom panels show simulated TD trajectories, sampled from the distributions of DCT coefficients given in Table 1. For reference, the red line denoting the direct trajectory is drawn in the lower panels as well. Note that, even though the mean of the DCT coefficients are based on the straight line, the stochastic simulations are non-linear. This is because the distributions over DCT coefficients define the same range of variability as is present in the TD trajectories, which are also not linear.

Figure 6

Top panels show raw TD data for / ϕ uzoku/ (left) and / ϕ usoku/ (right). The bottom panels show simulated trajectories from the vowel deletion hypothesis.



Focusing on the bottom panels of Figure 6 observe the “accidental” vowels. That is, some of the trajectories belonging to the targetless hypothesis have an increase in height in the middle of the trajectory. If observed in isolation, these tokens could be misinterpreted as arising from active high vowel constrictions. The presence of “accidental” vowels underscores an important point about evaluating phonological hypothesis on the basis of phonetic data and the role of stochastic modelling. It is crucial to consider the level of variability in the data. In the case at hand, we find that amongst tokens actualizing linear interpolation there are some that show a rise in tongue dorsum height at approximately the point in time that we would expect the vocal tract to be under control of a vowel gesture. The presence of accidental vowels indicates that the level of normal variation that characterizes fluent production of a native language is on the order of magnitude of the presence/absence of a vowel in one or two out of a dozen or so tokens.

At this point, it is now possible to statistically adjudicate between three of our four hypotheses. In asking whether the vowel is targetless in the upper right panel of Figure 6, we are essentially asking whether the rise of the TD in the middle of the trajectory is greater than can be expected by chance. We have defined chance for / ϕ usoku/ as our targetless trajectory in the bottom right corner of Figure 5. Using sparse representation of the data, as in Table 1, we can statistically compare / ϕ usoku/ to / ϕ uzoku/ to see if the TD trajectory in the devoiced vowel is significantly different from the TD trajectory in the voiced vowel. This constitutes a direct statistical test of H1, the hypothesis that devoiced vowels are the same as the voiced counterparts. A significant difference

would falsify H1, leaving us with H2 and H3, i.e., that the lingual gesture in devoiced vowels is either reduced (H2) or deleted (H3). Further, we can compare the TD trajectory in / ϕ usoku/ to the simulated targetless trajectory to see if the data are significantly different from linear interpolation. This constitutes a statistical test of H3. A significant difference would leave us with H2, as the only viable alternative. However, this method does not allow us to test H4, the variable targetlessness hypothesis. The reason is that in evaluating statistical significance in this way, we are testing whether the tokens as a group are different, which involves the implicit assumption of phonological homogeneity across tokens of a word. The next computational tool we introduce alleviates this assumption. We note, however, that if it can be ensured that a phonological process is not optional, then DCT together with Micro-prosodic sampling should suffice to provide a rigorous assessment of linear interpolation.

3.0 Bayesian Classification

A number of phonological processes are optional. Capturing the variability requires a probabilistic phonological model (Anttila 1997; Boersma and Hayes 2001; Coetzee and Kawahara 2013). In these models, phonetic reduction and variable deletion (or even variable targetlessness, as in our case) are completely different scenarios. The latter requires stochastic interpretations of constraint rankings (or rules); the former requires continuous phonological representations of some sort (e.g., Smolensky et al. 2014). We therefore incorporate a third component to our computational toolkit, a naïve Bayesian classifier, which will allow us to analyze the data token-by-token without committing to the assumption that the surface phonological form of a word is uniform and invariant.

The Bayesian classifier assigns the probability of category membership. Importantly, it does so for each test token separately. For the case at hand, we use the DCT representation (four coefficients) as input to the classifier. The output is the probability of whether the articulatory target in that token comes from the “target present” category or the “targetless” category. The naïve Bayesian classifier coefficients that describe the data are statistically independent. The basic formula is provided in (2). The output of the formula is the posterior probability of targetlessness, i.e., the term $p(T|Co_1, Co_2, Co_3, Co_4)$, which designates the probability of a targetless articulation given the four DCT coefficients. The alternative (to a targetless articulation) is that there is a vowel target present. The posterior probability of a vowel target is calculated from a prior probability of targetlessness and the probability of the DCT values given the category. The prior probability of targetlessness is the term $p(T)$. The probability of the DCT values given the category is the term $p(Co_1, Co_2, Co_3, Co_4|T)$. This is calculated on the basis of the training data and is normalized by a third term, the probability of the DCT coefficients in the whole data set: $p(Co_1, Co_2, Co_3, Co_4)$.

(3) Formula for Naïve Bayesian Classifier

$$p(T|Co_1, Co_2, Co_3, Co_4) = \frac{p(T) p(Co_1, Co_2, Co_3, Co_4|T)}{p(Co_1, Co_2, Co_3, Co_4)}$$

where

T = targetless, target present
 Co_1 = 1st DCT Coefficient
 Co_2 = 2nd DCT Coefficient
 Co_3 = 3rd DCT Coefficient
 Co_4 = 4th DCT Coefficient

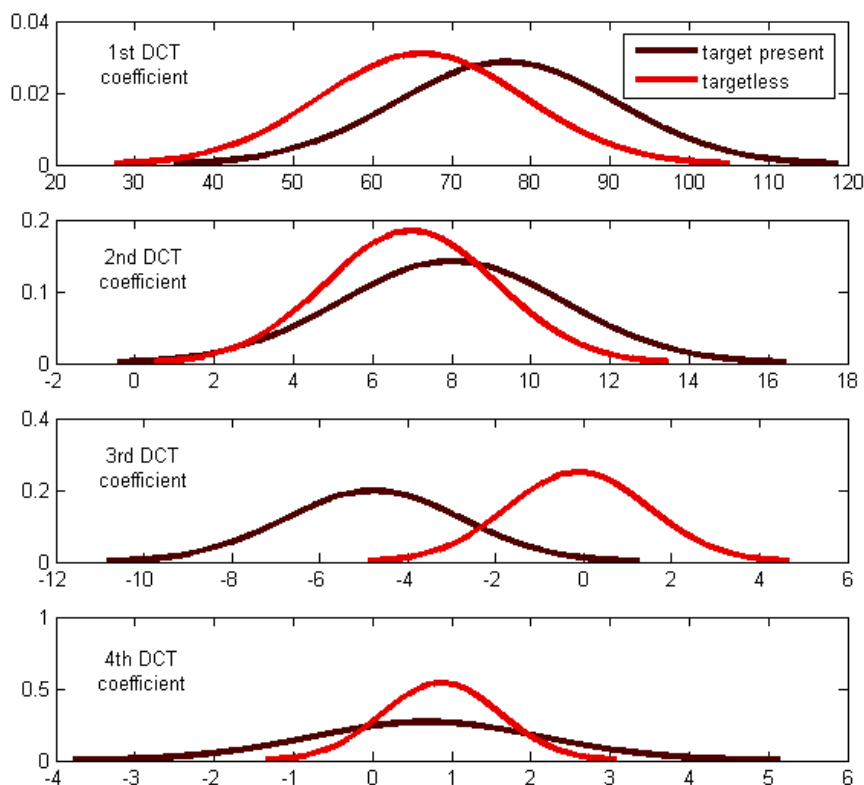
In this particular case, we are concerned with assessing the four hypotheses in (1) on the basis of the phonetic signal. To give each hypothesis equal weight, we assign equal *a priori* probabilities to the categories target present, H1 in (1), and target absent, H3 in (1) hypotheses. Thus, $p(T)$ is set to

.5. The other hypotheses, vowel reduction (H2) and variable targetlessness (H4), can also be evaluated on the basis of posterior probability patterns, as we illustrate below (Figure 8).

We trained the Bayesian classifier on two sets of four DCT coefficients. The “target present” data came from DCT coefficients fit to / ϕ uzoku/. The “target absent” data came from DCT coefficients fit to the linear interpolation trajectory from /e/ to /o/ (as in Figure 6, bottom). Since we have set the prior probability to even odds of targetlessness, it is the probability of each DCT coefficient given the presence/absence of a TD height target that dictates posterior probabilities. To visualize this factor, Figure 7 compares Probability Distribution Functions (PDF)’s across hypotheses, target present vs. target absent, for each DCT coefficient. The black lines show PDFs over the “targetless” hypothesis; the red lines show the “target present” hypothesis. As can be seen from the figure, the PDF’s of the 1st, 2nd, and 4th DCT coefficients are heavily overlapped. The main difference between the presence and absence of a vowel is found in the PDF of the 3rd DCT coefficient. This is expected since the 3rd coefficient dictates the magnitude of the TD rise between /e/ and /o/ vowels (see Figure 5). Thus, the parameters of the Bayesian classifier for / ϕ usoku/ give quantitative probabilistic form to the observations we have already made about the data.

Figure 7

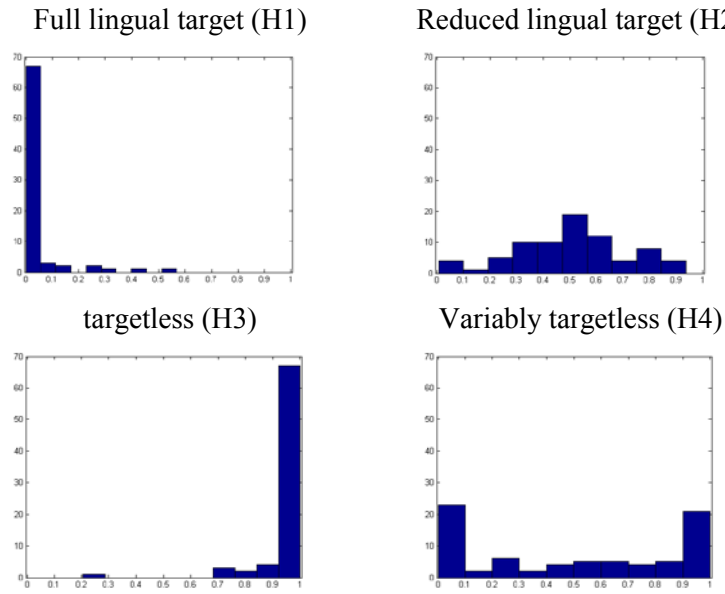
The probability distribution functions (PDFs) for DCT coefficients given the targetless hypothesis and the alternative target present hypothesis based on the data in Figure 9.



Four possible patterns, displayed as histograms over posterior deletion probabilities, are illustrated in Figure 8. These hypothetical results correspond to the four hypotheses laid out in (1). The histogram in the top left panel was obtained by submitting the / ϕ uzoku/ tokens (from 6 speakers) to the Bayesian classifier. As expected, most of these token have greater than .95 probability of containing a vowel, although there are a few tokens with lower probabilities. This pattern corresponds to H1, that the lingual gestures for voiced vowels are the same as for voiceless vowels. The histogram in the bottom left was obtained by submitting the same number of simulated “vowel absent” trajectories to the classifier. Again, as expected, most tokens have a .95 or greater probability of targetlessness, although there are a few tokens with lower probabilities and one “accidental vowel” which has a low-ish (.25) probability of deletion. This pattern corresponds to H3, that the lingual gestures of devoiced vowels are targetless. The third pattern, illustrated in the top right panel shows posterior probabilities for reduced vowels, H2. These were generated by stochastic sampling of DCT coefficients that were averaged between “target present” (H1) and “target absent” (H3) values. Thus, quite literally, the reduced vowel cases are intermediate trajectories between the fully articulated vowel and the targetless vowel. The fourth pattern, representing H4, is the variable deletion pattern. We created this pattern by averaging the targetless and the full vowel target tokens. In the following sections, we report methods for data collection and results of applying the tools described in this section to data collected from 6 speakers of Tokyo Japanese.

Figure 8

Four hypothetical posterior probability patterns. The vertical axis of each histogram shows posterior probabilities generated by the Bayesian classifier summarized in Figure 7.



Posterior probabilities of targetlessness

4.0 Electromagnetic Articulography Experiment

The experiment used Electromagnetic Articulography (EMA) (e.g., Hoole and Zierdt 2010; Perkell et al. 1992) to track the articulatory kinematics of devoiced high vowels in Japanese.

4.1 Speakers

Six native speakers of Tokyo Japanese (3 male) participated. Participants were aged between 19 and 22 years at the time of the study. They were all born in Tokyo, lived there at the time of their participation in the study, and had spent no more than 3 months outside of the Tokyo region.

4.2 Materials

The stimuli in the experiment consisted of words presented in the carrier phrase: *ookee* _____ *to itte* 'Ok, say _____'. The preceding word /*ooke*/ ending with [e] was selected so that the tongue would be in a non-high position at the start of the target word. A rise in tongue position from [e] to [u] would therefore suggest the presence of vowel target for [u]. To illustrate the computational approach we focus on the four words in Table 2. In these words, the target vowel [u] occurs in either a devoicing environment (left column) or a voiced environment (right column). In both contexts, /u/ is unaccented. These words were randomized in a list of 16 other words, 10 of which did not contain high vowels in a devoicing context. All words were randomly displayed within the carrier phrase in normal Japanese script. Participants were instructed to speak as if they were making a request of a friend. Each participant produced a total of 10-15 repetitions of each target word.

Devoiced vowels	Voiced vowel
ʃutaisei 主体性 'willingness'	ʃudaika 主題歌 'theme song'
φusoku 不足 'shortage'	φuzoku 付属 'enclosed'

Table 2: stimulus items

4.3 Equipment

We used an NDI Wave EMA system sampling at 100 Hz to capture articulatory movement. NDI wave 5DoF sensors were attached to three locations on the sagittal midline of the tongue, and on the lips, jaw (below the lower incisor), nasion and left/right mastoids. The tongue dorsum sensor is the focus of our analysis here. It was the most posterior of the three sensors on the tongue, attached as far

back as was comfortable for the participant. Acoustic data were recorded simultaneously at 22 KHz with a Schoeps MK 41S supercardioid microphone.

4.4 Post-processing

Following the main recording session, we also recorded the bite plane of each participant by having them hold a rigid object, with three 5DoF sensors attached to it, between their teeth. Head movements were corrected computationally after data collection with reference to three sensors on the head, the left/right mastoid and nasion sensors, and the three sensors on the bite plane. The head corrected data was rotated so that the origin of the spatial coordinates corresponds to the occlusal plane at the front teeth.

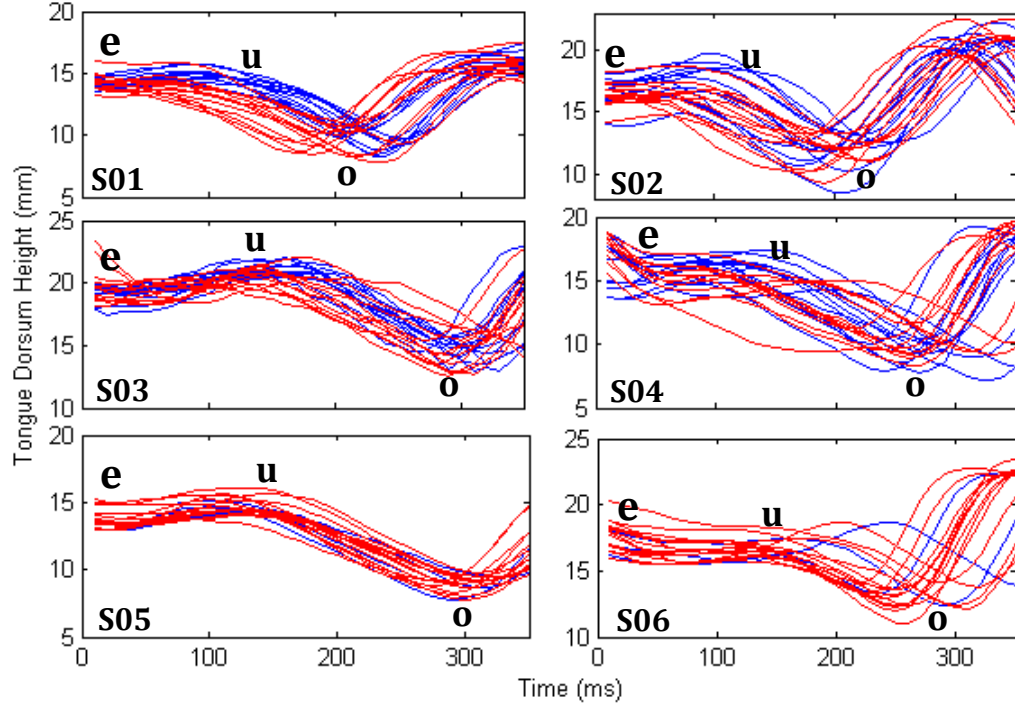
5.0 Results

5.1 Analysis based on DCT and Micro-prosodic sampling

Figure 9 shows TD height trajectory in / ϕ usoku/ and / ϕ uzoku/ from six speakers. The data used to illustrate the modelling approach in section 2 comes from S01. Red lines show change in tongue dorsum height over time for / ϕ usoku/; blue lines show / ϕ uzoku/. The tongue height trajectories begin with the /e/ of the carrier phrase and continue for 350 ms. The dip in the trajectories corresponds to lowering of the TD for the vowel /o/. This is followed by a rise of the TD for /k/ at the ends of the trajectories. Only the portion of the trajectory spanning from /e/ to /o/ was included in subsequent analysis.

Figure 9

Change in Tongue Dorsum (TD) height (y -axis) over time (x -axis) for / ϕ usoku/ (redlines) and / ϕ uzoku/ (blue lines).



Following the method introduced in section 2, we fit DCT coefficients (section 2.1) to each TD height trajectory and defined a targetless trajectory (section 2.2). We then compared DCT components by MANOVA. For each speaker, we evaluated the effect of voicing on TD trajectory as well as differences between the actual trajectories and the targetless trajectory. Results are summarized in Table 3. Since the targetless trajectory is stochastically sampled, statistical comparisons vary depending on the particular sample evaluated. To ensure stable (and replicable) statistical results, below we report the average across 10 independent simulations of the targetless trajectory. Since we conducted these analyses for each subject and each pair of items separately, we adjust the alpha level to correct for multiple comparisons. The Bonferroni corrected alpha is $\alpha = 0.00138$ ($.05/36$), where 36 is the total number of comparisons: 6 speakers X 2 item pairs X 3 comparisons per item pair.

In Table 3, significant differences are indicated by asterisk. Of the six participants, four produced reliable differences between the vowels in / ϕ usoku/ and / ϕ uzoku/. For all six participants, the voiced vowel in / ϕ uzoku/ was significantly different from the targetless trajectory. Of the four speakers who produced / ϕ usoku/ and / ϕ uzoku/ differently, only one speaker, S04, produced the devoiced vowel in / ϕ usoku/ consistent with the targetless trajectory. For the other three, / ϕ usoku/ was significantly different from the targetless trajectory. Thus, only one of the six speakers, S04, produced the devoiced vowel with a tongue dorsum height trajectory that could not be distinguished from the targetless trajectory.

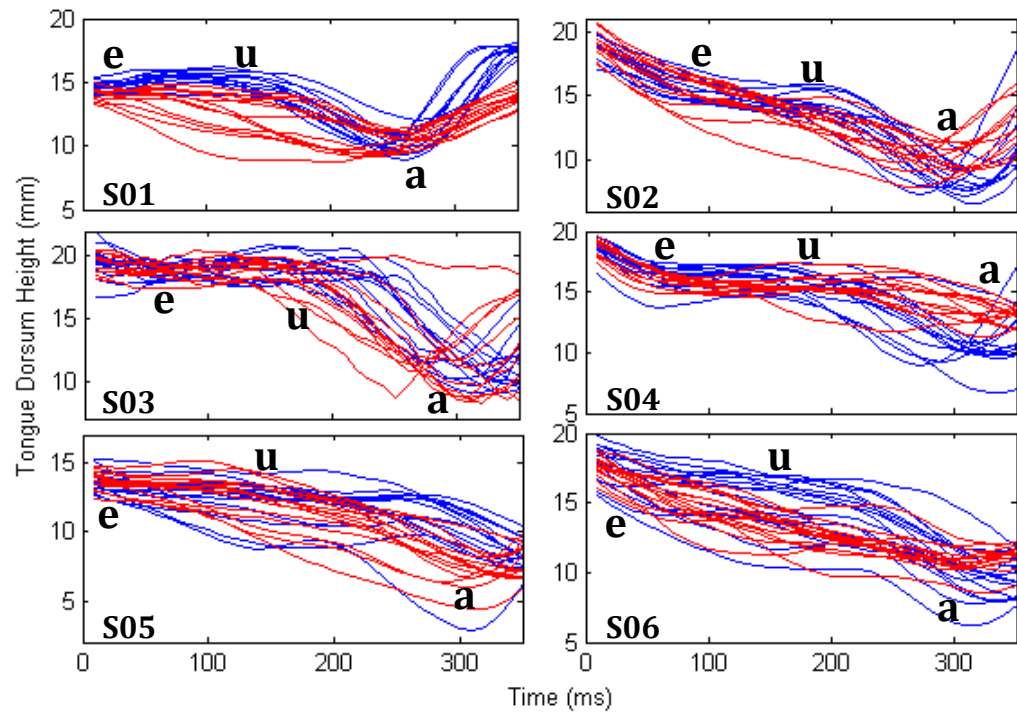
Subject	Comparison	<i>df</i>	<i>F</i>	<i>p</i>	lambda
S01	ϕusoku ~ ϕuzoku	21	22.9	0.0001*	0.2798
	ϕuzoku ~null	21	30.2	0.0000*	0.1912
	ϕusoku ~null	21	18.2	0.0012*	0.3641
S02	ϕusoku ~ ϕuzoku	25	20.8	0.0004*	0.3891
	ϕuzoku ~null	25	45.6	0.0000*	0.1289
	ϕusoku ~null	25	25.1	0.0002*	0.3270
S03	ϕusoku ~ ϕuzoku	23	26.8	0.0000*	0.2624
	ϕuzoku ~null	23	47.5	0.0000*	0.0948
	ϕusoku ~null	23	27.8	0.0002*	0.2602
S04	ϕusoku ~ ϕuzoku	23	23.3	0.0001*	0.3114
	ϕuzoku ~null	23	28.4	0.0000*	0.2459
	ϕusoku ~null	23	5.8	0.3138	0.7559
S05	ϕusoku ~ ϕuzoku	27	0.2	0.9953	0.9917
	ϕuzoku ~null	27	51.1	0.0000*	0.1204
	ϕusoku ~null	27	54.5	0.0000*	0.1047
S06	ϕusoku ~ ϕuzoku	29	0.2	0.9971	0.9940
	ϕuzoku ~null	29	32.9	0.0000*	0.2854
	ϕusoku ~null	29	25.2	0.0002*	0.3832

Table 3: MANOVA results for /ϕusoku/ and /ϕuzoku/ for each speaker.

Figure 10 shows the trajectory of tongue dorsum height for another pair of words, /ʃutaisei/ and /ʃudaika/. The red lines show the word containing the devoiced vowel, /ʃutaisei/; the blue lines show the comparison word, /ʃudaika/, which contains a voiced /u/. For all six speakers, the TD trajectory is somewhat lower for the devoiced vowel (red lines) than for the voiced vowel (blue lines). Moreover, for several speakers the red lines take an almost linear trajectory between flanking vowels /e/ and /a/. To assess the statistical significance of these trends we again followed the methods outlined in Section 2. We fitted DCT components to each trajectory, simulated a targetless trajectory and compared these via MANOVA. The results are summarized in Table 4.

Figure 10

Change in TD height (y -axis) over time (x -axis) for /futaisei/ (red lines) and /fudaika/ (blue lines).



As shown in Table 4, all six speakers produced /futaisei/ and /fudaika/ with significantly different tongue height trajectories. Moreover, of the six speakers, only one, S05, produced /futaisei/ so that it was significantly different from the targetless trajectory. For completeness, we also note that one speaker, S02, who did not produce a difference between /futaisei/ and the targetless trajectory also did not produce a difference between /fudaika/ and the targetless trajectory that was significant after Bonferroni correction ($p = 0.009$ where $\alpha = 0.001$).

Subject	Comparison	<i>df</i>	<i>F</i>	<i>p</i>	lambda
S01	<u>f</u> utaisei ~ <u>f</u> udaika	21	22.0	0.0002*	0.2949
	<u>f</u> udaika ~null	21	47.9	0.0000*	0.0703
	<u>f</u> utaisei ~null	21	3.5	0.5050	0.8250
S02	<u>f</u> utaisei ~ <u>f</u> udaika	25	20.3	0.0004*	0.3980
	shudaika~null	25	16.9	0.0097	0.4739
	<u>f</u> utaisei ~null	25	8.2	0.1826	0.6994
S03	<u>f</u> utaisei ~ <u>f</u> udaika	23	17.9	0.0013*	0.4078
	<u>f</u> udaika ~null	23	42.1	0.0000*	0.1232
	<u>f</u> utaisei ~null	23	16.9	0.0054	0.4357
S04	<u>f</u> utaisei ~ <u>f</u> udaika	23	52.3	0.0000*	0.0732
	<u>f</u> udaika ~null	23	27.2	0.0000*	0.2598
	<u>f</u> utaisei ~null	23	11.1	0.0378	0.5770
S05	<u>f</u> utaisei ~ <u>f</u> udaika	27	22.8	0.0001*	0.3861
	<u>f</u> udaika ~null	27	21.0	0.0012*	0.4218
	<u>f</u> utaisei ~null	27	41.5	0.0000*	0.1796
S06	<u>f</u> utaisei ~ <u>f</u> udaika	29	49.3	0.0000*	0.1504
	<u>f</u> udaika ~null	29	17.8	0.0013*	0.5041
	<u>f</u> utaisei ~null	29	12.0	0.0176	0.6312

Table 4: MANOVA results for /futaisei/ and /fudaika/ for each speaker.

To summarize, most speakers showed significant differences in tongue height trajectories between voiced and voiceless vowels. This result allows us to rule out the possibility that devoiced vowels are produced with the same lingual articulatory gestures as voiced vowels, H1 in (1). With respect to targetlessness, the statistical evaluation indicates that /u/ is sometimes targetless. One speaker, S04, produced / ϕ usoku/ and five speakers produced /futaisei/ without a clear height target. Thus, a conclusion based upon this analysis is that devoiced vowels are often reduced and sometimes even produced without a target. Moreover, we could divide speakers, based on this analysis into three groups. For / ϕ usoku/, there are speakers who produce /u/ without a height target (S04), those who reduce /u/ (S01, S02, S03) and those who produce full vowels (S05, S06). For /futaisei/, there are two groups: those who reduce (S05) and those who produce a targetless /u/ (S01, S02, S03, S04, S06). We caution, however, that analysis by MANOVA treats as a homogenous group all tokens of / ϕ usoku/ and /futaisei/ for a given speaker. If there is within-speaker optionality, then this assumption is not justified. We therefore conclude with an approach that allows us to evaluate targetlessness on a token by token basis. This approach will evaluate H4 in (1), optional targetlessness, and offer an additional angle on the other hypotheses, H1-H3.

5.1 Bayesian classification

We submitted each token of / ϕ usoku/ shown in Figure 9 and each token of /futaisei/ in Figure 10 to a Bayesian classifier as described in section 2.3. Recall that, as illustrated in Figure 8, all four hypotheses in (1) can be expressed as patterns of posterior probabilities, the output of the Bayesian classifier. For easy comparison to Figure 8, we have summarized the posterior probabilities as histograms.

Figure 11 provides a histogram of posterior probabilities for / ϕ usoku/. The left panel aggregates across speakers. The right panel provides a breakdown by speaker. The pattern is remarkably clear. There are two modes in the probabilities. One of them is around 0.05 probability of targetlessness; the other is around .95 probability of targetlessness. In fact, there are very few tokens at all that have intermediate probabilities, i.e., a token which we could call phonetically reduced. Across the six speakers, rather, it seems like /u/ in / ϕ usoku/ is optionally targetless. The breakdown of individual speakers in the right panel helps us to make sense of the MANOVA results. Recall that

speakers S01, S02, S03, showed significant differences between / ϕ usoku/ and / ϕ uzoku/ as well as between / ϕ usoku/ and the deletion trajectory. From the right panel of Figure 10, it is clear why. These speakers optionally produce the vowel without a height target. Speaker S04 does so most often. The main difference between speakers S01 through S04, therefore, is not a difference between phonetic reduction and phonological deletion but rather in the frequency with which the vowel is targetless. The other speakers, S05 and S06, produced no tokens classed as targetless. S05 was particularly remarkable in that all 11 tokens had a targetless probability of less than .05, and 10 of them had a probability less than .01.

Figure 11
Posterior deletion probabilities for 77 tokens of / ϕ usoku/ from 6 speakers (the TD trajectories shown in Figure 9). The left panel aggregates across speakers; the right panel shows probabilities by speaker.

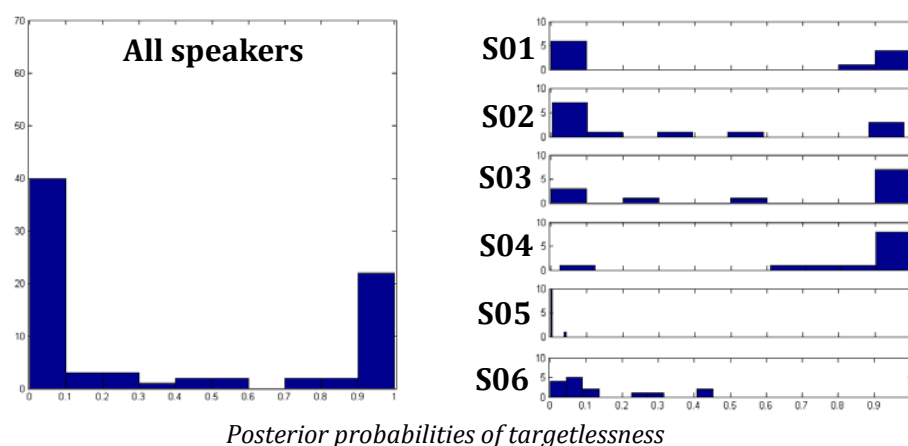
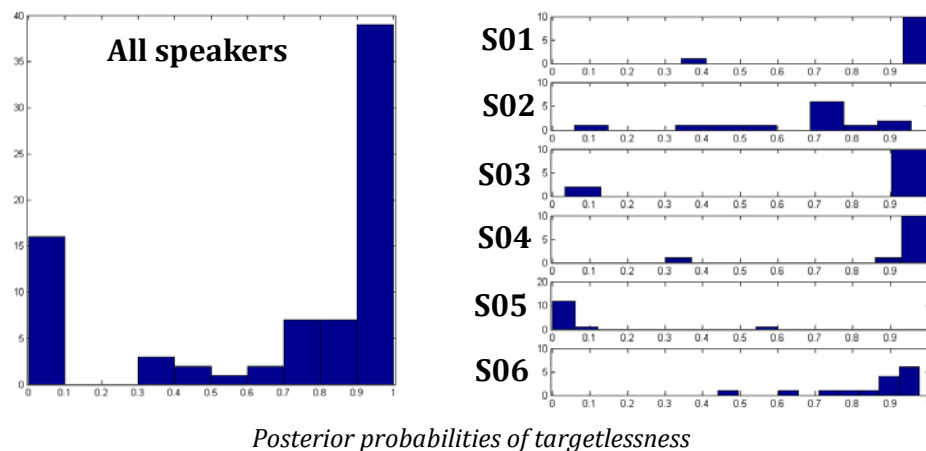


Figure 12

Posterior deletion probabilities for 77 tokens of /ɸutaisei/ from 6 speakers (the TD trajectories shown in Figure 10).



The Bayesian classification provides converging evidence for some of the conclusions based on MANOVAs but also provides additional insight. Both analyses indicate that targetlessness is more common in /ɸutaisei/ than in /ɸusoku/. This is true for the aggregate data as well as for each subject individually. Table 5 highlights this point, providing average targetless probabilities by speaker for each word. Although targetlessness varies across subjects, all subjects show a higher probability of targetlessness in /ɸutaisei/ than in /ɸusoku/. This goes as well for S05, who has a low probability of targetlessness in both words. Although the MANOVA analysis also indicated that targetlessness was more common in /ɸutaisei/ than in /ɸusoku/, Bayesian classification reveals that this holds across all speakers individually as well. Thus, while the overall probability of targetlessness seems to be a matter of personal preference, relative probabilities of targetlessness are shared across speakers.

	Targetless probability	
Subject	/ɸutaisei/	/ɸusoku/
S01	0.9385	0.4590
S02	0.6664	0.4646
S03	0.8234	0.7499
S04	0.9338	0.8269
S05	0.0591	0.0064
S06	0.8476	0.0968

Table 5: average probability of targetlessness by speaker and by word

Another new insight gained from Bayesian classification is the status of phonetic reduction, i.e., H2 in (1). When comparing DCT components via MANOVA, we found that four out of six speakers showed significant differences between voiced (/ɸuzoku/) and voiceless (/ɸusoku/) contexts. Of the four speakers that showed a difference between /ɸusoku/ and /ɸuzoku/, three of them also showed a significant difference between /ɸusoku/ and the vowel deletion trajectory. Since the productions of /ɸusoku/ are different, as a group, from /ɸuzoku/ and also, as a group, from linear interpolation, we might be tempted to conclude that the vowels are reduced (but not targetless). The Bayesian classification reveals that this conclusion is unwarranted. Rather, a group of /ɸusoku/ tokens from a single speaker may be different from both the full vowel trajectory in /ɸuzoku/ and the targetless trajectory because it contains a mix of full vowel and targetless tokens. The Bayesian classification revealed that this is indeed the case for /ɸusoku/. Production of a lingual target in devoiced vowels in Tokyo Japanese is optional but phonetic reduction is rare. Tokens are either

produced with a full vowel, similar to the voiced context, or with no lingual vowel target at all, as in the linear interpolation assumed for underspecified segments.

6.0 Discussion

6.1 Summary

We have illustrated three computational tools which, put together, allow rigorous assessment of the linear interpolation hypothesis. The general strategy is to express competing phonological hypotheses (including the linear interpolation hypothesis) in the dimensions of the data, including realistic levels of variability, and then to classify the phonetic data according to phonological structure. DCT compresses the phonetic data into a small number of phonologically relevant parameters that preserve phonetic detail. We defined linear interpolation in this DCT frequency space. Stochastic sampling from distributions over DCT coefficients enabled simulation of competing phonological hypotheses (target present vs. target absent) with the level of phonetic variability observed in the data, a step we refer to as Micro-prosodic sampling. Finally, we trained a Bayesian classifier to assign probabilities of targetlessness to new data. We have illustrated the method using tongue dorsum movements produced by speakers of Tokyo Japanese. Based on existing literature, we motivated four possible hypotheses about lingual articulatory targets and demonstrated step by step how our computational approach can adjudicate between them.

6.2 What we've learned about high vowel devoicing

Results for Tokyo Japanese indicate that the lingual articulatory gesture of devoiced vowels is rarely reduced, despite the fact that, given the devoicing, it can have only negligible auditory consequences. There are, however, two distinct phonetic outcomes for devoiced vowels. They can be produced with or without a vowel height target. This result supports H4 in (1), the hypothesis that devoiced vowels are optionally targetless.

Another interesting aspect of the results is that the probability of vowel targetlessness varied systematically across words. For all speakers, the probability of producing a vowel without a height target was higher for /ʃutaisei/ than for /ϕusoku/. Considering C_1 (/ʃ/ vs. /ϕ/) only, the result is somewhat surprising, but it could be due to resulting consonant cluster phonotactics. Deletion of /u/ in /ʃutaisei/ would give rise to a fricative-stop cluster, [ʃt], which may be a better surface form than the fricative-fricative cluster [ϕs] resulting from /u/ deletion in /ϕusoku/. If we assume that a syllable boundary remains between these surface consonants, a preference for fricative-stop clusters over fricative-fricative clusters is consistent with syllable contact laws (Vennemann 1988; Murray and Vennemann 1983). Since there is a greater distance in sonority between the offset of one syllable and the onset of the next, [ʃ.t] is marked relative to [ϕ.s] (Gouskova 2004). It is not clear exactly what other facts of Japanese, if any, motivate this fine-grained grammatical preference, although similar types of patterns have been observed in the production and perception of unfamiliar consonant clusters (Berent et al. 2009; Berent et al. 2007; Davidson and Shaw 2012; Shaw and Davidson 2011; Davidson et al. 2007).

Consistency across speakers in the relative targetlessness of /ʃutaisei/ and /ϕusoku/ resembles other well-studied cases of phonological variation, such as t/d deletion, in which grammatical influences remain constant even as overall deletion rates vary across speakers (Guy 1997; Coetzee and Kawahara 2013). We are planning additional experiments to further evaluate the effect of surface consonant cluster as well as other linguistic factors on targetless probabilities. Thus, although the final word on grammatical factors conditioning targetlessness must wait for additional data, it is clear that the computational approach we have developed can bring order to phonetic data, revealing phonological factors conditioning continuous movement dynamics with clarity and quantitative rigor.

6.3 Comparison with other approaches

Our approach differs from other quantitative attempts to assess phonological hypotheses, including targetlessness, on the basis of phonetic data. To highlight the uniqueness points, we briefly summarize past approaches, which can be divided into four categories: (1) heuristic use of phonetics

(2) statistical comparison of two samples of phonetic data (3) predicting one part of the phonetic signal from another (4) hypothesis testing by simulation.

The first approach, heuristic use of phonetics, involves drawing some conclusion about phonological form on the bases of visual inspection of the phonetic signal. Phonetic heuristics have played an important role in foundational work in laboratory phonology (Keating 1988; Cohn 1993). Phonetic heuristics can be extremely useful in augmenting transcriptions or auditory impressions of phonological form, particularly from field workers who are non-native speakers of the target language. However, phonetic heuristic may also break down. They are sometimes too sensitive and sometimes not sensitive enough. Consider, for example, a common phonetic heuristic for a vowel between stop consonants: “a period of voicing...with formant structure containing a visible second formant that ended with abrupt lowering of intensity at the onset of the second stop (Davidson 2010).” Application of this heuristic to Tashlhiyt Berber, for example, greatly overestimates the frequency of vowels in the language (Ridouane 2008). A Berber word like /t-bdg/ “it is wet” contains no vowels in the phonological representation but is normally pronounced with three periods of voicing that would meet the above-stated phonetic heuristic (Fougeron and Ridouane 2011). In this case, the phonetic heuristic is too sensitive. On the other hand, English words such as *support*, which contain two phonological vowels, are sometimes produced with just one phonetic segment meeting the above heuristic for a vowel (Davidson 2006b). In this case, the phonetic heuristic is not sensitive enough. Heuristics breakdown because they do not capture the full range of phonetic signals consistent with phonological form.

An alternative to visual inspection of the phonetic signal is to statistically compare one or more phonetic dimensions in two groups of words hypothesized to differ in phonological structure. A wide range of statistical tools have been deployed to this end. For example, Smoothing Spline ANOVAs can be used to compare populations of splines (Gu 2013), such as tongue shapes or the tongue height trajectories analyzed here, and have been applied to a range of phonetic signals (Davidson 2006a). However, a significant difference between two populations of signals does not necessarily indicate the nature of the phonological difference. As our case study demonstrates, the same word can be produced with different phonological specifications. Populations of signals can therefore be different not because they actuate different phonological structures but because they actuate different mixtures of phonological structures (also Shaw and Davidson 2011). Alternatively, populations of signals can differ due to phonetic factors. For example, Shaw et al. (to appear) demonstrate that tongue height in Mandarin Chinese vowel production varies across tones. Despite the common claim that tones and vowels are phonologically independent (Yip 2002), there are dependencies between laryngeal and supralaryngeal articulation that result in small but statistically significant sub-categorical differences in lingual articulation for the same vowel produce with different tones. Thus, statistical differences between surface measurements offer no guarantee of a categorical phonological difference between samples. As with heuristic use of phonetics, statistical comparison of continuous dimensions can be over-sensitive, picking up differences that do not correspond to phonological structure.

A third approach is to use one part of the phonetic signal to predict another. Specifically in the area of targetlessness, Pierrehumbert and Beckman (1988: 37-38) rely on this approach to argue for sparse tonal specifications in Japanese unaccented words. They argue that in unaccented words, only the first two syllables are specified as LH, and there is a general decline toward the L tone in the next Accentual Phrase. They show that the longer the duration between H and L, the shallower the slope. In this case, the duration between H and L tones is used to predict the slope of the f₀ fall. The relationship between these phonetic variables constitutes an argument for the tonal targetlessness of intervening syllables. This specific correlation, since it requires manipulating the duration of the hypothesized “targetless” material, does not easily generalize to other cases of targetless, e.g., nasal specification of English vowels, etc, but conceptually similar approaches have been applied to other arguments for targetlessness. Browman and Goldstein (1992) used a multiple regression framework to assess whether English schwa contains an articulatory target. They reasoned that schwa could be

claimed to be targetless in sequences such as /pV₁pəpV₂p/ if the spatial position of the articulators could be reliably predicted by the flanking vowels in a two parameter (one coefficient for each flanking vowel) linear regression model (also Lammert et al. 2014). They argued that schwa in such words has a target of its own, since regression models with an intercept term, representing the mean height of the signal, tended to outperform models informed only by flanking vowel positions.

In modelling contextual effects on English schwa, Browman and Goldstein (1992) also deploy what we would describe as a fourth type of approach. They simulated phonetic data from various phonological hypotheses, including targetlessness, and compare the simulated data to the experimental data. They find a qualitative match between simulated data and experimental data when English schwa is specified in the model with a neutral vowel target and overlapped in time with the following vowel, a result that converges nicely with the regression analysis described above, and makes different predictions than the targetless specification (particularly in high vowel environments). Browman and Goldstein (1992) explore several possible phonological configurations by specifying gestural scores by hand and examining the phonetic consequences. More recent simulations derive gestural scores from coupled oscillators (Goldstein et al. 2009; Saltzman et al. 2008; Nam 2007) or posit coordination topologies isomorphic with syllable structure while fitting lower level parameters to the data (Shaw and Gafos 2015, 2010; Gafos et al. 2014). The computational toolkit we have introduced belongs to this fourth type of approach, which assesses competing phonological hypotheses computationally, by simulating those hypotheses in the physical dimensions of phonetic data.

In comparison with other models instantiating the fourth class of approaches described above, our toolkit is, in some ways, more bottom-up, requiring fewer theoretical commitments and also fewer researcher degrees of freedom.² First, the parameters in the model, i.e., the values of the DCT coefficients, are determined by the data, according to the algorithm in (1). Second, our approach does not privilege particular points in time, i.e., “magic moments” (Mücke et al. 2014), as having greater phonological relevance than others. In many of the studies described above, specific moments in time are selected for analysis. For example, Browman and Goldstein (1992) and Shaw et al. (to appear) select, by automatic algorithm, a specific point in time to represent the spatial position of a vowel. Regardless of the algorithm, whether based on displacement of articulators, formant values, peak velocity, the temporal midpoint of voicing, etc., “target” selection introduces a researcher degree of freedom into the analysis. Our toolkit alleviates the necessity of picking points in time. This aspect of our approach is particularly useful for addressing the presence/absence of a target, as it is problematic to choose a point in time corresponding to a target that might not be there. Thus, our approach makes the presence/absence of targets a largely empirical question which can be addressed with phonetic data. The one assumption that we have adopted is that targetlessness corresponds to linear interpolation in the phonetics. Beyond this, since the parameters capturing phonetic signal modulation are fit to the data quantitatively, the bottom-up approach remains compatible with most higher level theories of phonological representation, including dynamically defined gestural units, as in Articulatory Phonology (e.g., Browman and Goldstein 1990).

6.4 Broader application of the toolkit

Although the computational toolkit we have assembled to assess linear interpolation takes continuous phonetic data as input, the results for devoiced vowels in Japanese are remarkably categorical. Most tokens are either produced without a vowel target or with a full vowel target. The toolkit itself does not dictate such categorical outcomes (see, e.g., Figure 8). With respect to high vowel devoicing in Japanese, the categorical nature of the variation, as revealed by application of the toolkit, and its interaction with other grammatical factors suggests a distinctly phonological character

² “Researcher degrees of freedom” refers to flexibility in methods of data collection, reporting and analysis that affect statistical assessment of a hypothesis (see, e.g., Simmons et al. 2011).

to the phenomenon. We are optimistic about the prospects of applying these computational tools to a wider variety of phenomenon and curious about the extent to which other cases of “phonetic reduction” are actually manifestations of optional phonological processes. One domain within which the current toolkit may be particularly applicable is intonation. As mentioned in the introduction, the issue of underspecification (targetlessness) is particularly important in intonation (see, e.g., Xu et al. (2015) vs. Arvaniti and Ladd (2015)). Since intonation comes with much natural variability, just like our articulatory data, application of these tools to the tonal underspecification hypothesis may prove to be highly informative.³

7.0 Conclusion

We have assembled a set of computational tools to assess the presence vs. absence of a phonological element on the basis of phonetic data. From end to end, the set of tools, consisting of Discrete Cosine Transform, Micro-prosodic sampling, and Bayesian Classification, does so without requiring explicit labelling of particular phonetic landmarks. In this sense, the toolkit can be productively deployed as a phonological feature detector. We demonstrated the approach with analysis of EMA recordings of voiced and devoiced vowels in Tokyo Japanese, contributing to a debate about whether devoiced vowels are specified for lingual articulatory targets. Analyzed within the computational framework described here, these data elucidated some previously unknown aspects of the pattern, including its highly categorical nature and phonological conditions under which devoiced vowels also lack lingual articulatory targets.

Largely data-driven, adaptable to a range of phonetic signals, and compatible with a broad spectrum of representational frameworks in phonology, we anticipate that the computational toolkit can be widely deployed to link hypotheses about the specification (or non-specification) of phonological elements, including features, gestures, and tones, to phonetic data.

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³ Relatedly, we note that the Fujisaki model of intonation (Fujisaki 1983) is somewhat akin to our DCT analysis in that it has two components: the “phrasal control mechanism” determines the overall intonational contour (as our 2nd DCT coefficient) and “the accent control mechanism” realizes local f0 movement due to pitch accent (as our 3rd and 4th coefficients). There is independent observation that tones may differ in terms of how large the domain of their influence is; for example, a tone that is attached to a phrasal node may influence a sequence of syllables, whereas a tone that is associated with a particular syllable may have a more localized effect (Pierrehumbert and Beckman 1988: 72-75). The former type of tones would be expressed in terms of lower DCT coefficients. In this sense, we already have a fairly good understanding of the linguistic interpretations of resulting DCT coefficients.

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